

Convolutional Neural Networks I

Semester 2, 2025

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Outline

- Image recognition
- Review of neural networks
- Convolutional neural networks

Learning outcomes

- Define image recognition as a computer vision problem
- Explain the differences between convolutional and fully-connected networks
- Implement convolutional layers

Image recognition

What is “recognition”?



“dog”

“shiba inu”

“Doge”

Image: <https://kabochan.blog.jp/archives/9733755.html>

What is “recognition”?

- In this section, we’ll define **image recognition** as category-level recognition of the whole image
- Category-level = group level
 - Groups may be more or less specific (“bird,” “duck,” “Australian wood duck”)
 - Different from **instance-level recognition**, recognising a specific individual
- Whole image = one label per image
 - Different from **detection** = locate object in image
 - Different from **segmentation** = label individual pixels

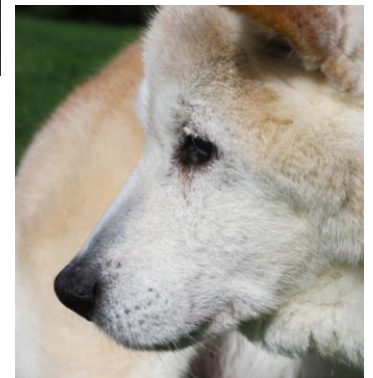
Why is recognition difficult?

- Inter-category similarity



Why is recognition difficult?

- Intra-category variability
 - Instances
 - Illumination
 - Scale
 - Viewpoint/pose
 - Background/occlusion

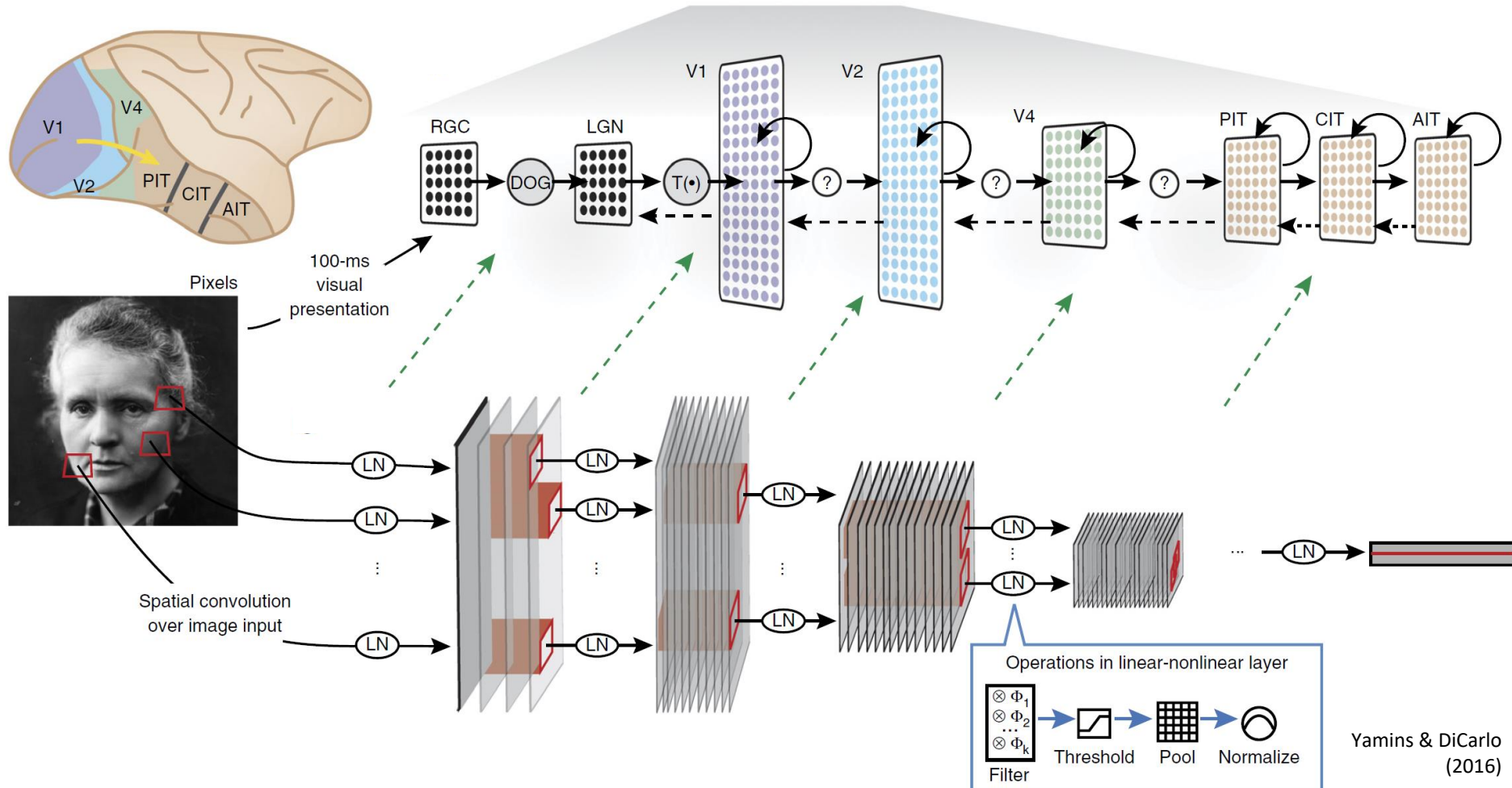


Goal of image recognition

Build a representation of the image that

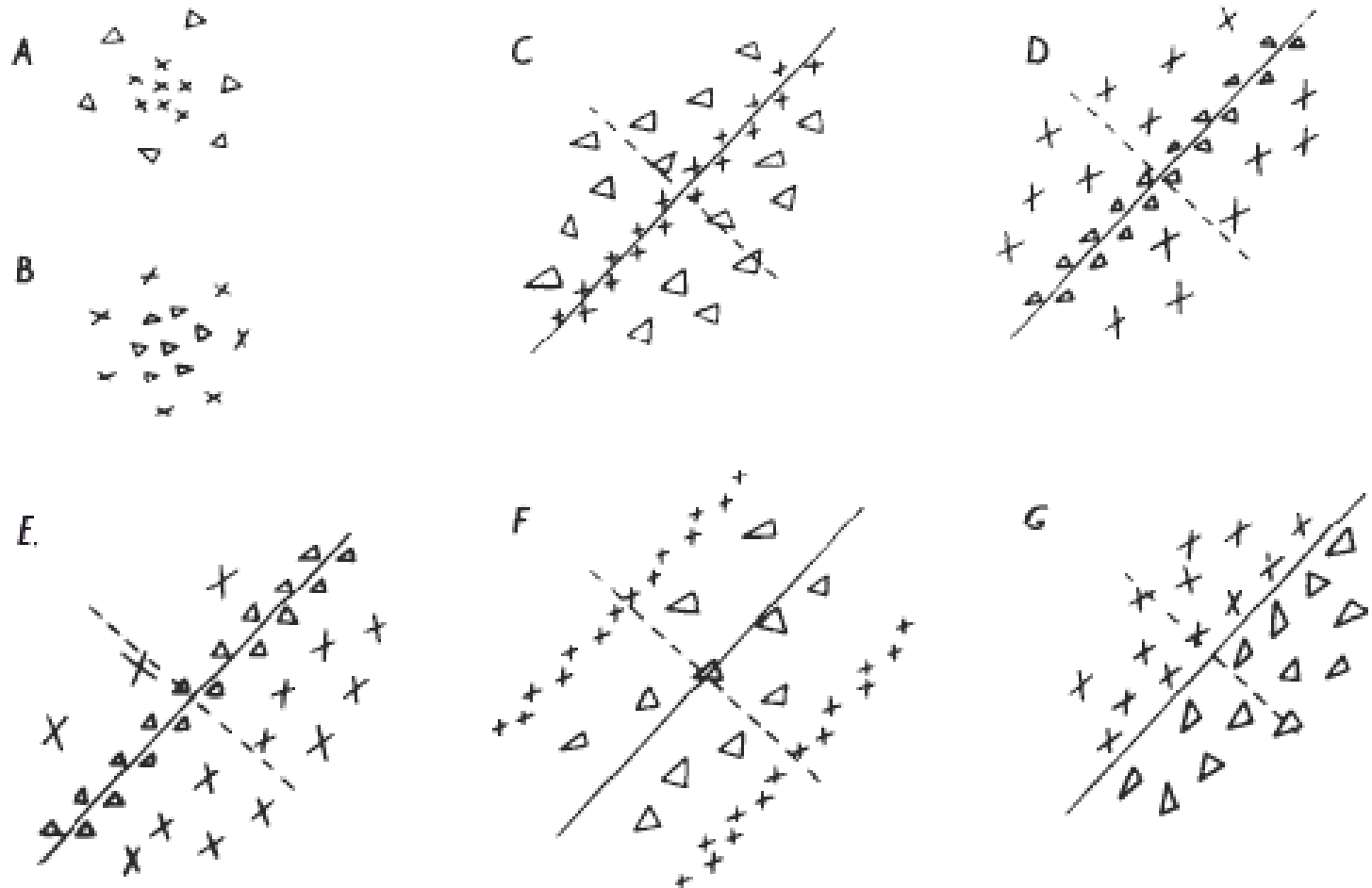
- a) distinguishes different categories,
- b) but is invariant (or tolerant) to variation within a category

Visual encoding

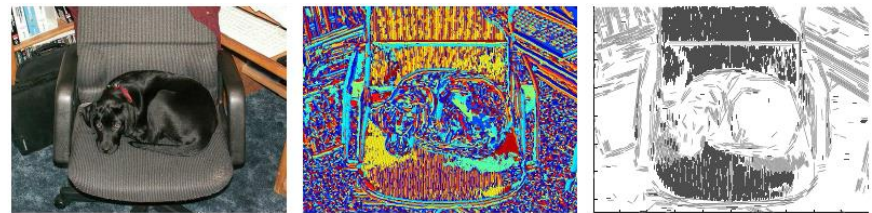
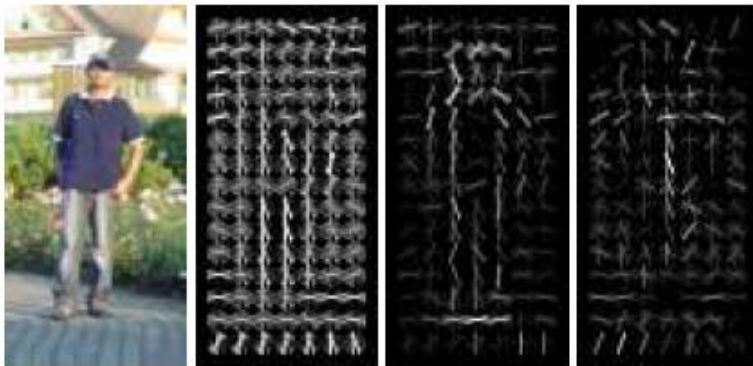
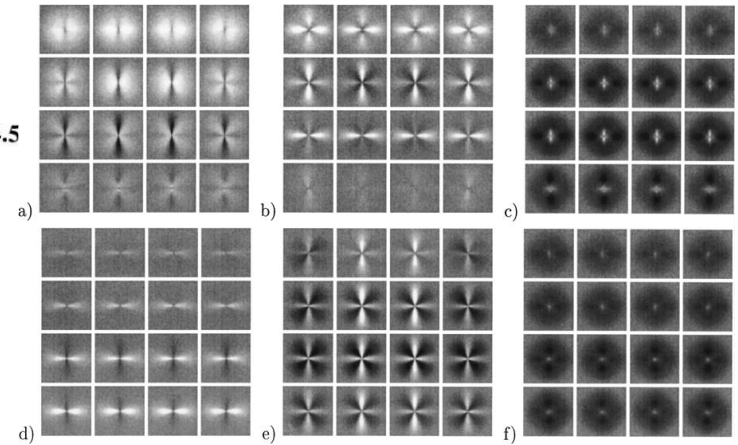
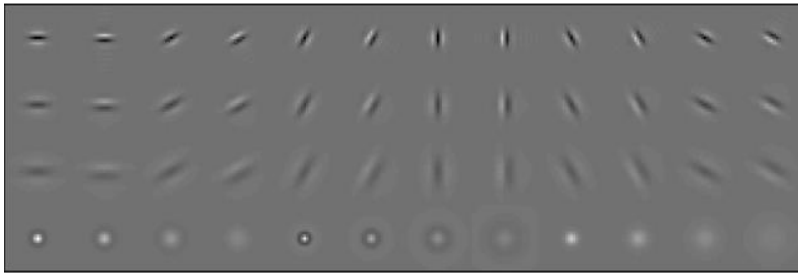
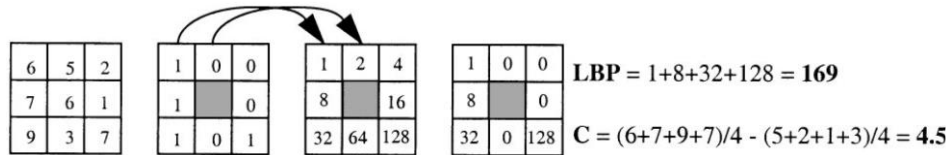


Yamins & DiCarlo (2016)

Visual encoding – early vision

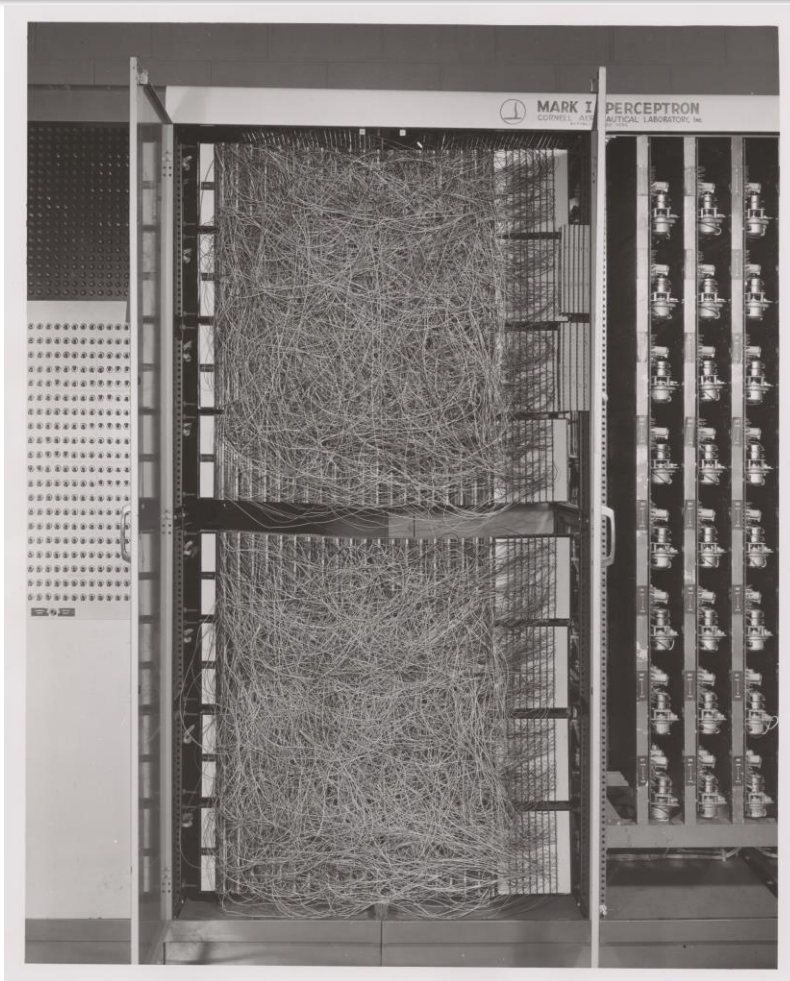


Hand-crafted features



Ojala & Pietikäinen (1999)
 Leung & Malik (2001)
 Oliva & Torralba (2001)
 Dalal & Triggs (2005)
 Vedaldi & Zisserman (2012)

Deep learning “revolution”



- Neural networks have been used for computer vision problems since 1950s
- But they only became the state-of-the-art around 2010
- What changed?

Cornell University News Service records, #4-3-15. Division of Rare and Manuscript Collections, Cornell University Library.
<https://digital.library.cornell.edu/catalog/ss:550351>

What changed?

- Graphics processing units (GPUs) for fast parallel computation
 - CUDA (Nvidia, 2007) – platform for parallel processing on GPU
- Algorithm improvements:
 - Unsupervised pretraining, ReLU activation function, regularisation (e.g., dropout)
- The internet
 - Massive amounts of text, photos, etc. *with human-annotated labels*

ImageNet

- Based on WordNet, a database of English words organised by concepts
- Class images collected online, manually cleaned by human annotators (2.5 years of annotation work)
- Over 5,000 classes, but commonly-used dataset includes just 1000 of the classes with most exemplars

ImageNet

[Main](#) [Instructions](#) [Unsure? Look up in Wikipedia](#) [Google](#) [\[Additional input \]](#) [No good photos? Have expertise? comments? Click here!](#)

[First time workers please click here for instructions.](#)

Click on the photos that contain the object or depict the concept of : **delta**: a low triangular area of alluvial deposits where a river divides before entering a larger body of water; "the Mississippi River delta"; "the Nile delta" .(PLEASE READ DEFINITION CAREFULLY)
Pick as many as possible. **PHOTOS ONLY, NO PAINTINGS, DRAWINGS, etc.** It's OK to have other objects, multiple instances, occlusion or text in the image.

Do not use back or forward button of your browser. OCCASIONALLY THERE MIGHT BE ADULT OR DISTURBING CONTENT.



Below are the photos you have selected FROM THIS PAGE ONLY (they will be saved when you navigate to other pages). Click to deselect.

[what's this?](#) [select all](#) [deselect all](#)

[<](#) page 1 of 6 [>](#)

[Submit](#)

PREVIEW MODE. TO WORK ON
THIS HIT, ACCEPT IT FIRST.

Large-scale image datasets

IMAGENET



<http://www.image-net.org/>

1.2 million images
1000 object classes

Open images dataset

<https://g.co/dataset/open-images>

9 million images

19,700 classes

YouTube-8M

<https://research.google.com/youtube8m/>

6 million videos

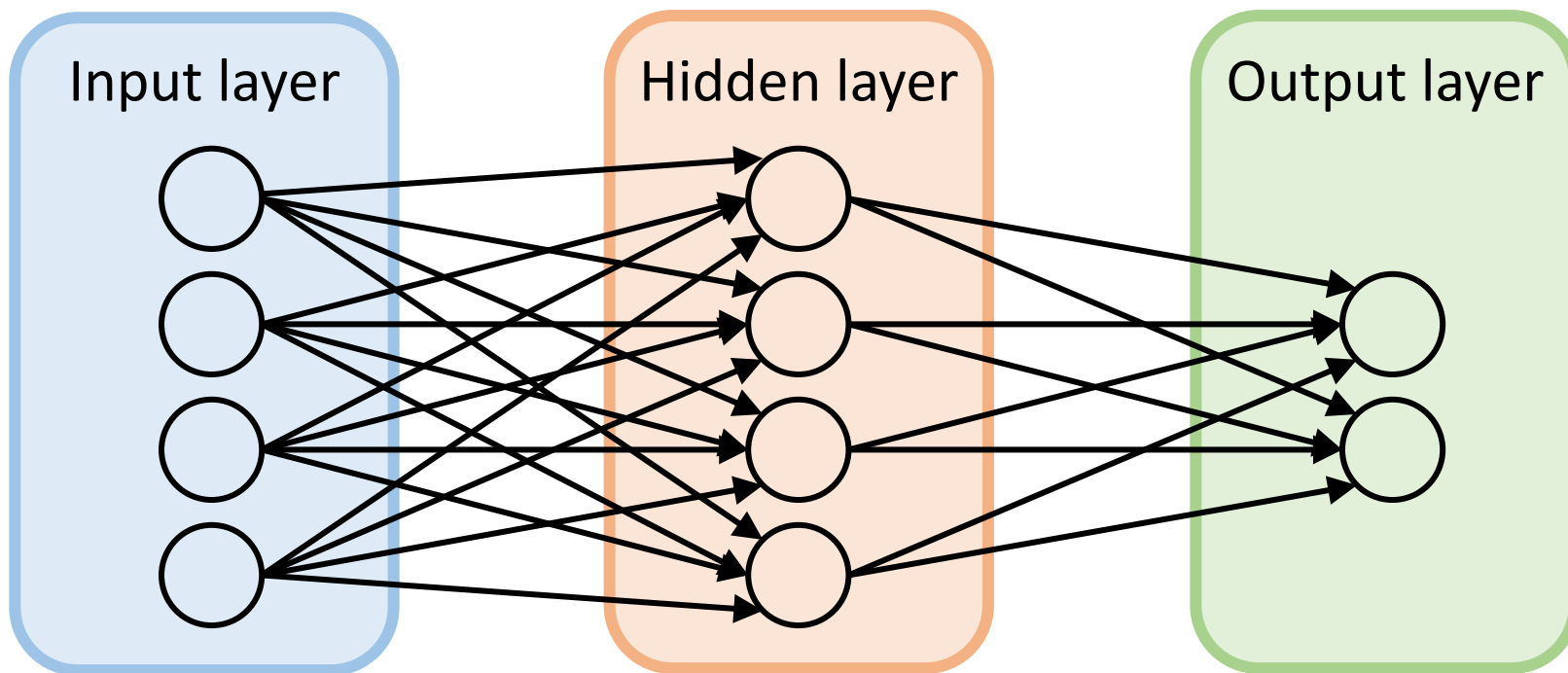
3800 object classes

Image recognition

- Supervised learning problem – map image to class label
- Pre-2010: small number of classes (order of 10-100), hand-crafted features
- 2010-now: “large scale” image recognition (order of 1000-10,000 classes), millions of images, deep-learned features

Neural network review

Neural networks / MLPs



MLP = multilayer perceptron

Neural networks

- Multiple layers of neurons working in parallel on the same input
- Each neuron on layer L receives input from all neurons on layer L-1 (**fully connected layer**) and produces one output
- Neuron's output is a weighted sum of the input, followed by a non-linear activation function

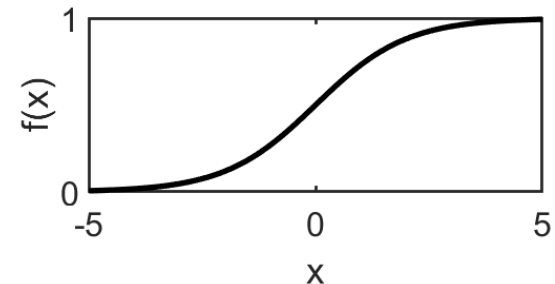
$$y = f \left(\left[\sum_j w_j x_j \right] + b \right) = f(\mathbf{w} \cdot \mathbf{x} + b)$$

Weights and bias learned from data

Non-linear activation function?

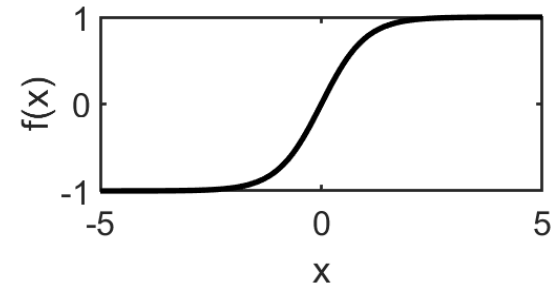
- (logistic) sigmoid (σ):

$$f(x) = \frac{1}{1 + e^{-x}}$$



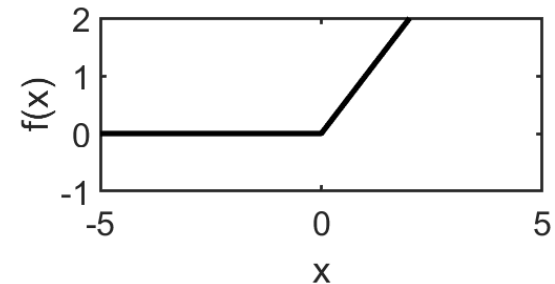
- hyperbolic tan (tanh):

$$f(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$



- rectified linear unit (ReLU):

$$f(x) = \max(0, x)$$



Neural networks

- Train through backpropagation (a form of gradient descent)
- Compute gradient of the loss function with respect to network parameters, starting with output layer and propagating to earlier layers, and adjust weights to reduce loss
- Learning rate is a free parameter
- Loss function usually based on difference between ground truth and prediction (supervised learning)

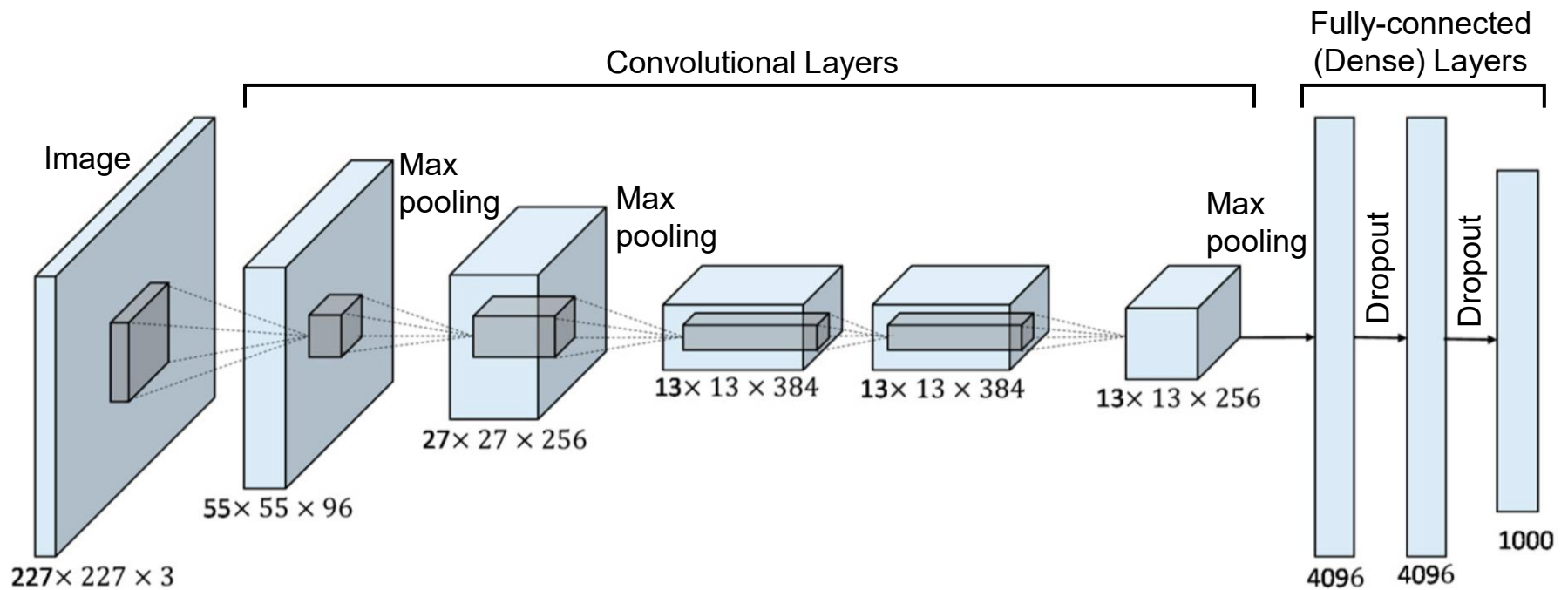
Neural networks

- Advantages
 - Universal approximator – able to approximate any continuous function on $\mathbb{R}^n$
 - Feature embedding – learns complex features
 - Parallelisable – within each layer, neurons are independent
- Disadvantages
 - Very large number of parameters – high memory/time/data requirements, prone to overfitting

Convolutional layers

Convolutional neural network

“AlexNet”: Krizhevsky, Sutskever, & Hinton (2012)



Why convolution?

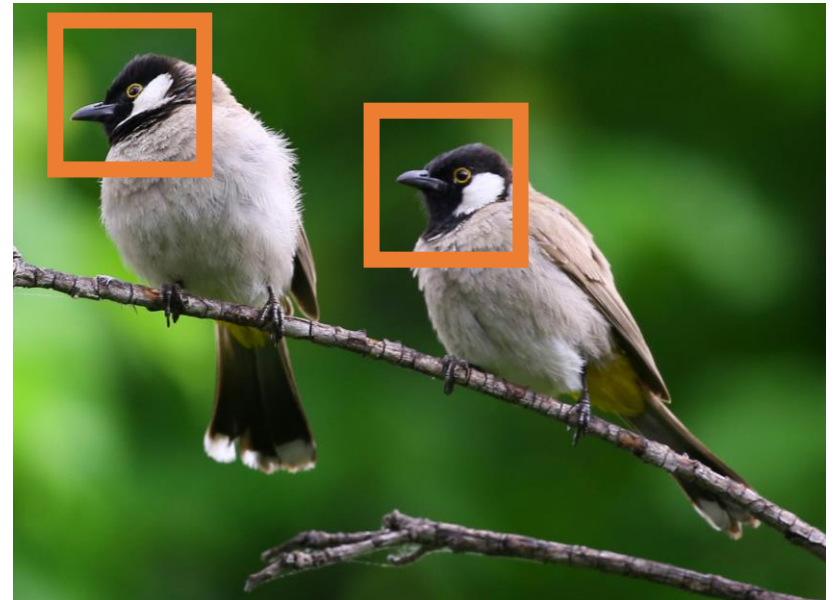
- Regular neural networks can be used for image recognition
- But **convolutional** neural networks are more common for large images
- Why?

Why convolution?

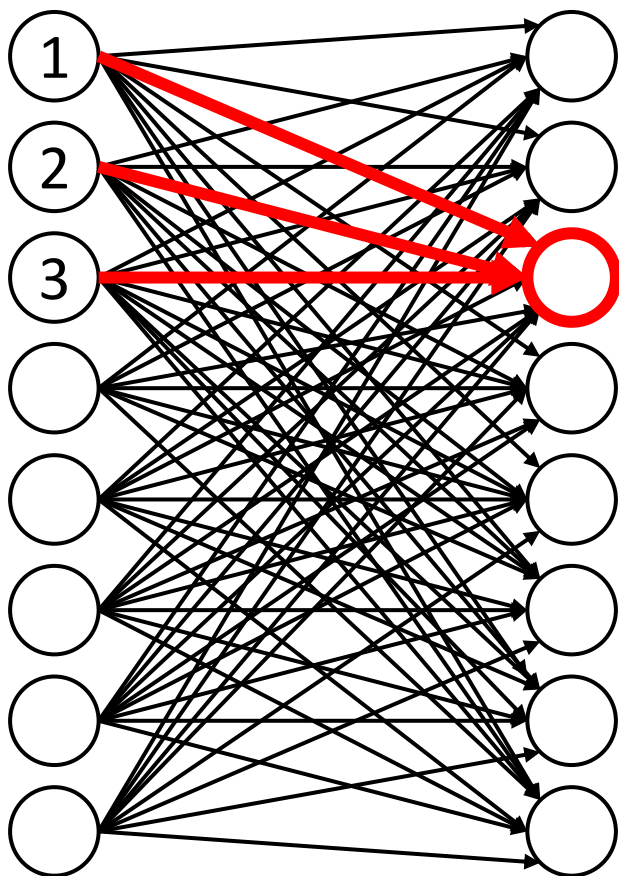
This is how i felt
while watching this
film. I loved it.
It was hilarious.

I loved it, it was
fun and moved
quickly, no boring
drawn out scenes.

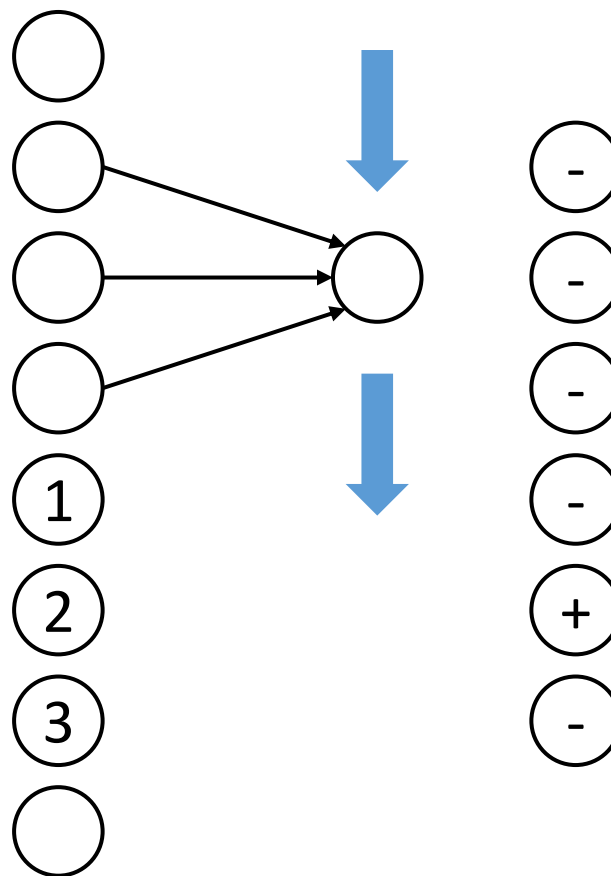
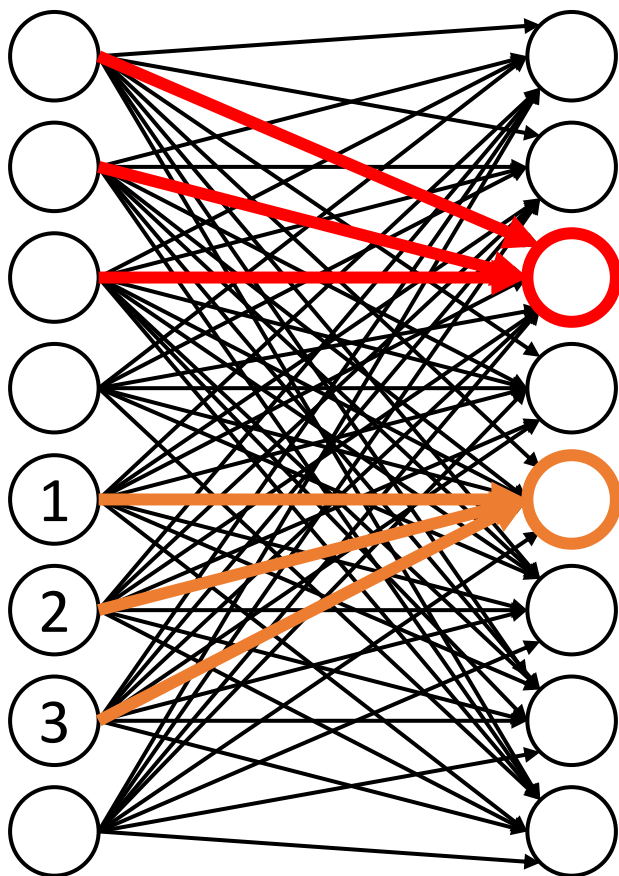
I just got it and
it is a great
movie!! I loved it!



Convolutional layer



Convolutional layer



Why convolution?

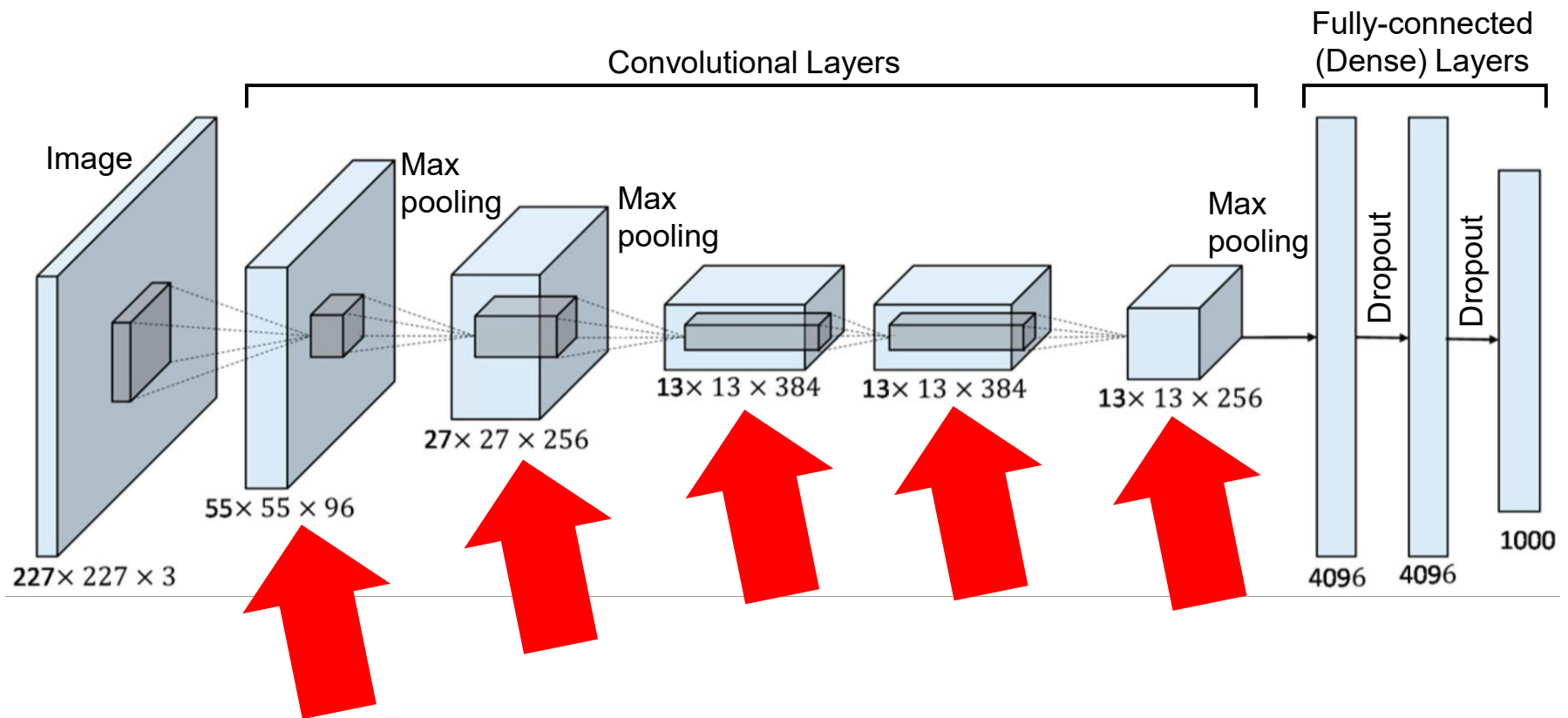
- Regular neural networks can be used for image recognition
- But **convolutional** neural networks are more common for large images
- Why?
 - More efficient learning of local, repeated patterns
 - However, limits what the network can learn

Convolutional layer

- Convolutions are defined by
 - A **kernel**, which is a matrix overlaid on the image and computes an element-wise product with the image pixels
 - A **stride** which defines how many positions in the image to advance the kernel on each iteration (stride = 1 means the kernel will operate on every pixel of the image)

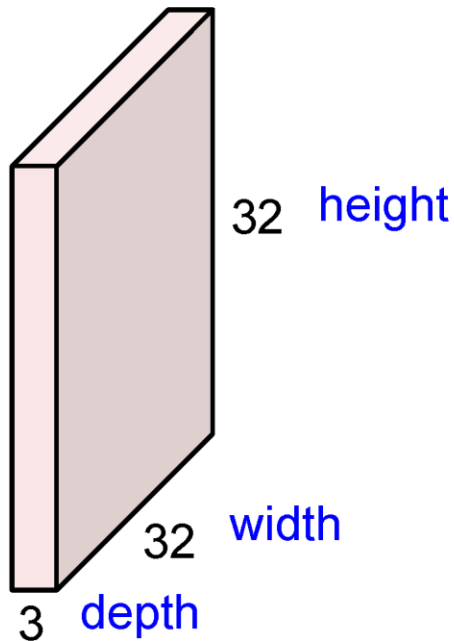
Convolutional layer

“AlexNet”: Krizhevsky, Sutskever, & Hinton (2012)



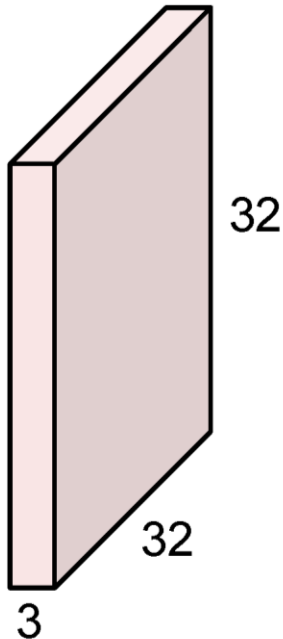
Convolutional layer

32x32x3 image -> preserve spatial structure



Convolutional layer

32x32x3 image



5x5x3 filter

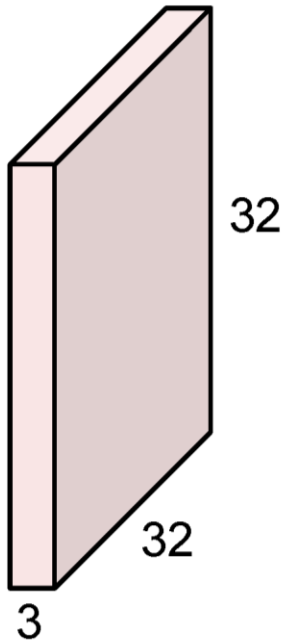


Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”

Technically, **most implementations do cross-correlation**, but we call it convolution anyway

Convolutional layer

32x32x3 image



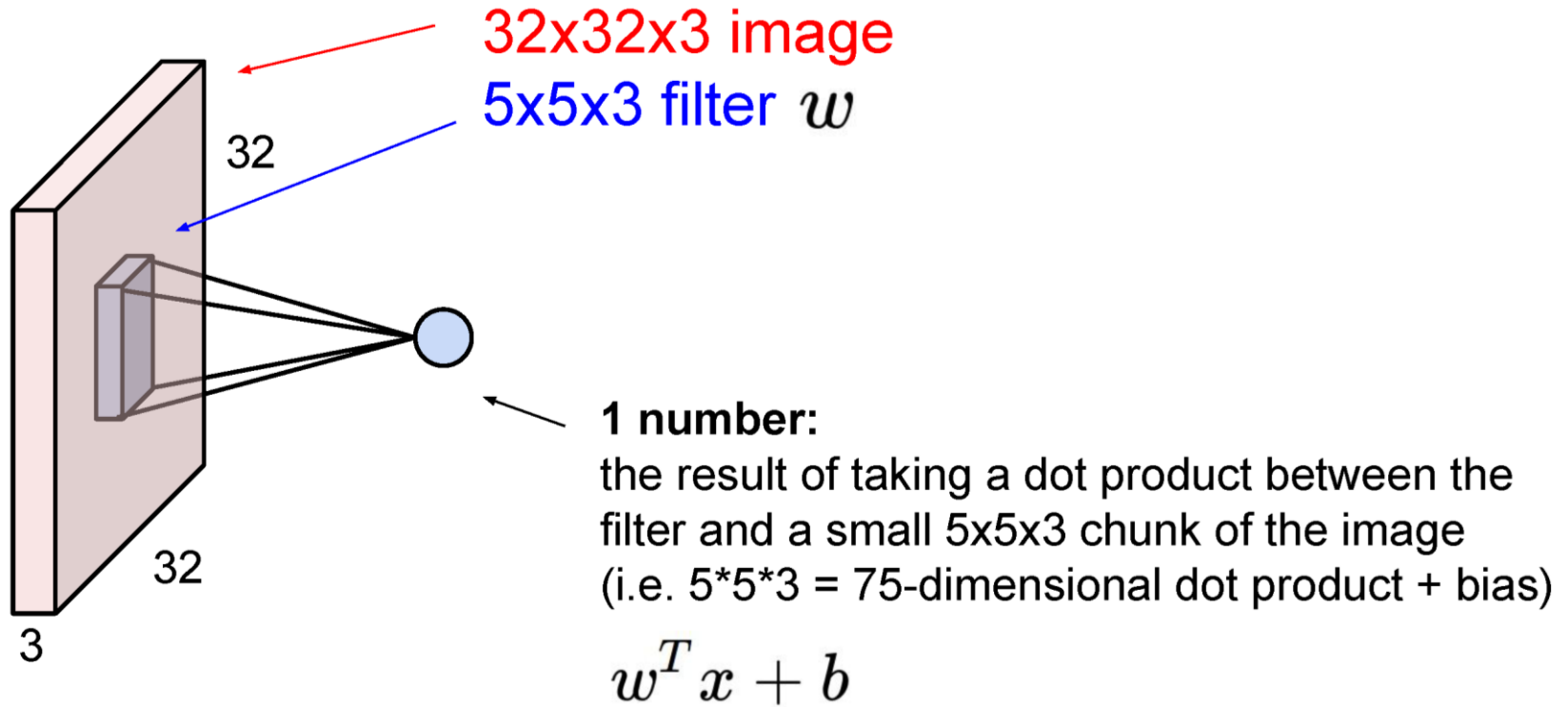
Colour image = 3 colour channels
Kernel also has 3 channels

5x5x3 filter

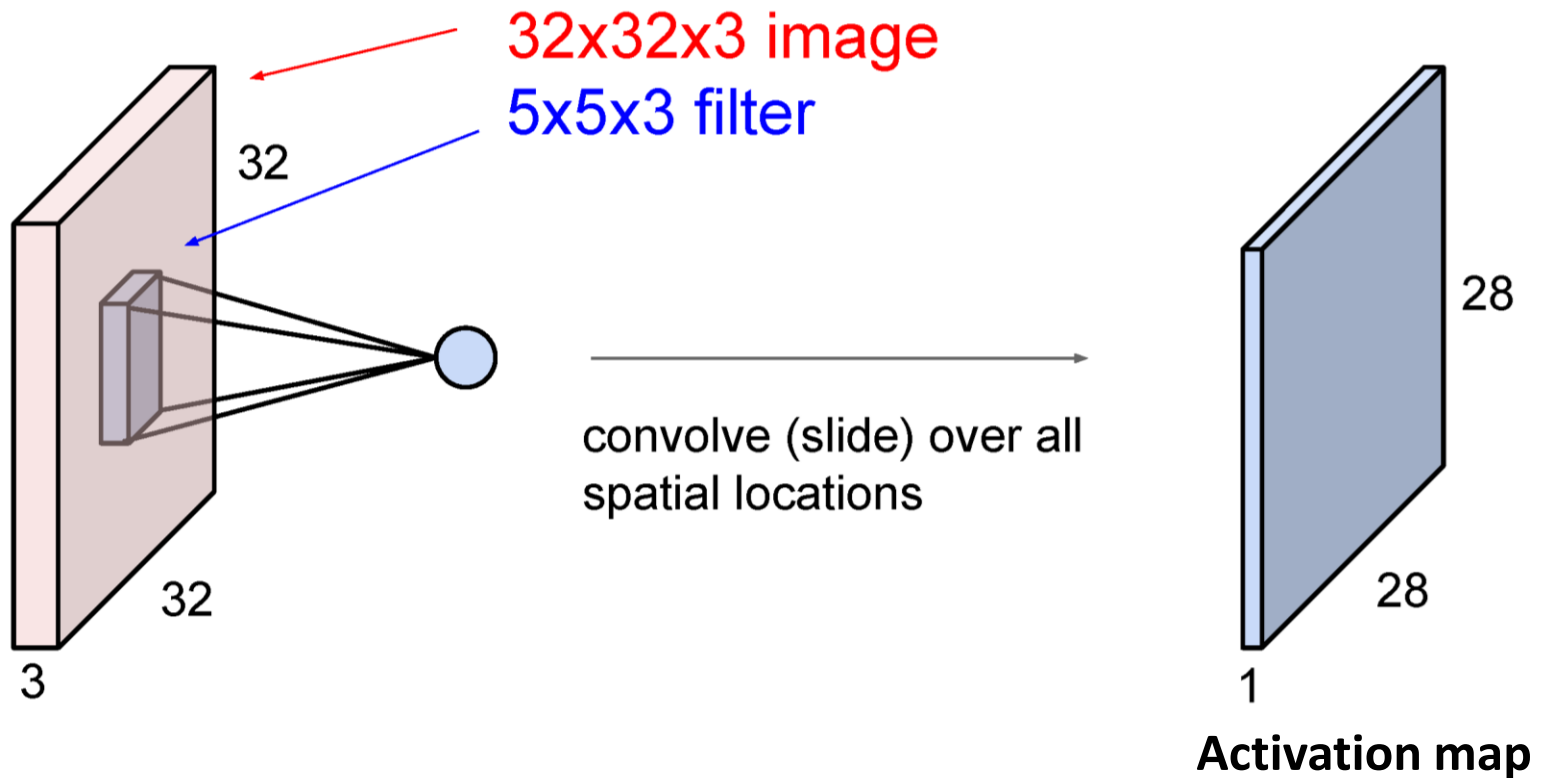


Convolve the filter with the image
i.e. “slide over the image spatially,
computing dot products”

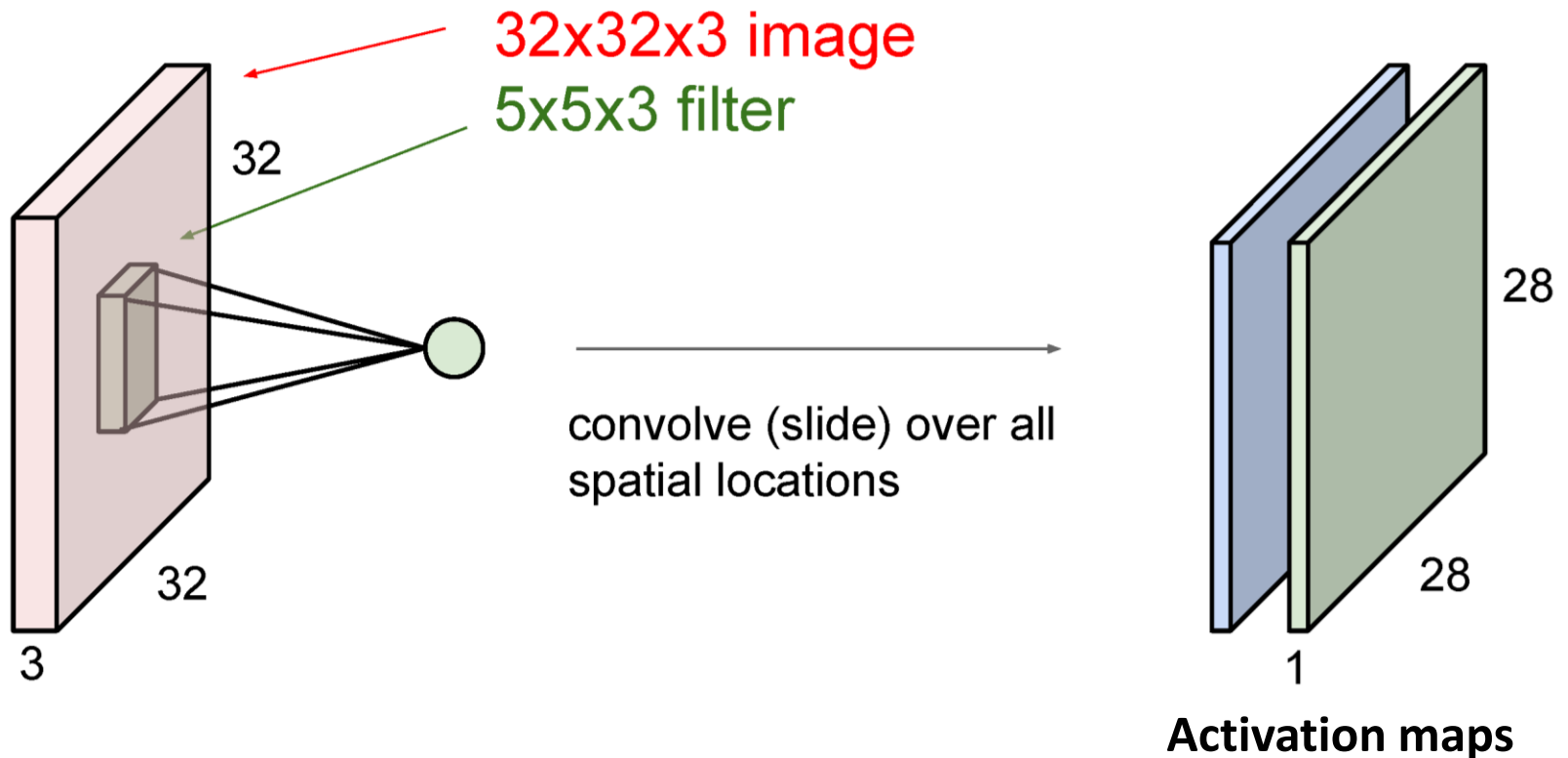
Convolutional layer



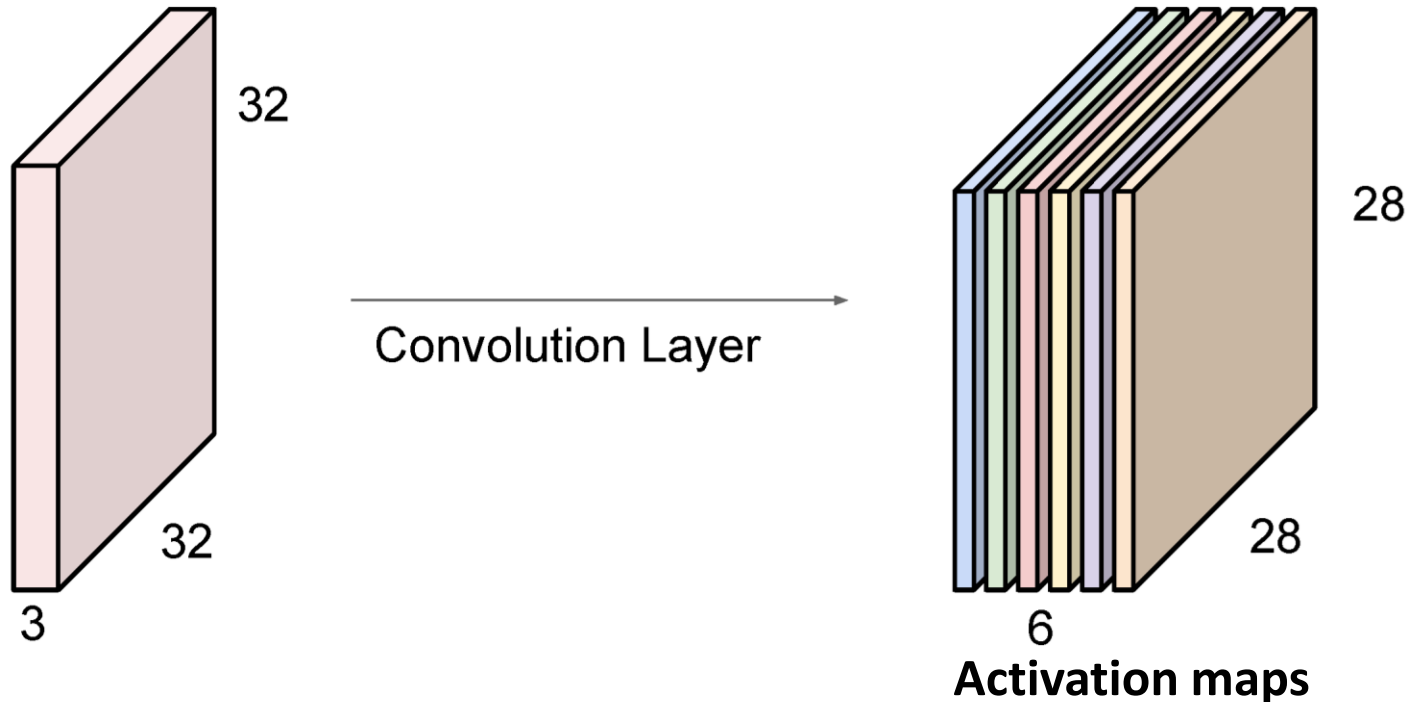
Convolutional layer



Convolutional layer



Convolutional layer



The stack of outputs makes a new 6-channel image, which will be the input to the next layer
6 channels = outputs of 6 different convolution kernels

Image: <http://cs231n.stanford.edu/>

Convolutional layer output



Input = image

Neuron 1 kernel:



Output:



Neuron 2 kernel:



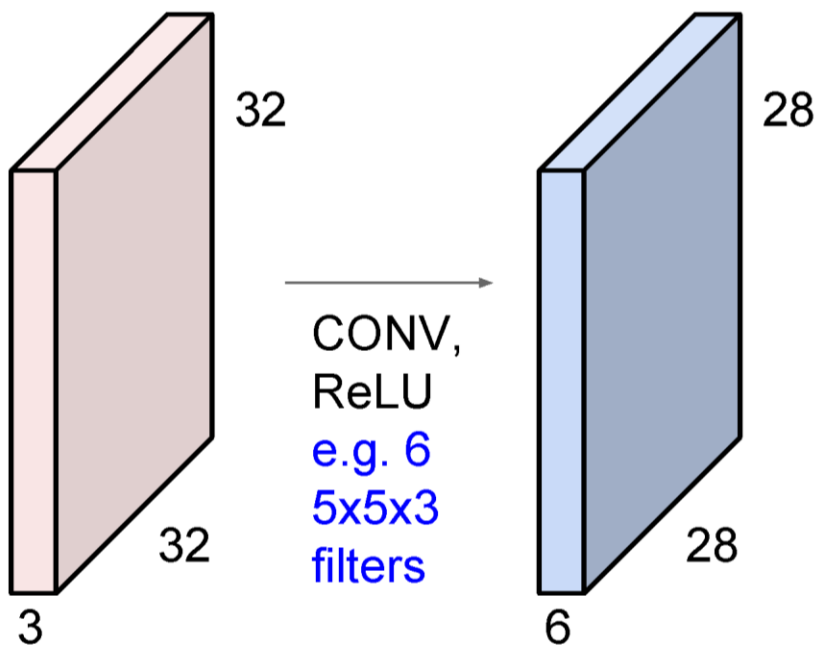
Output:



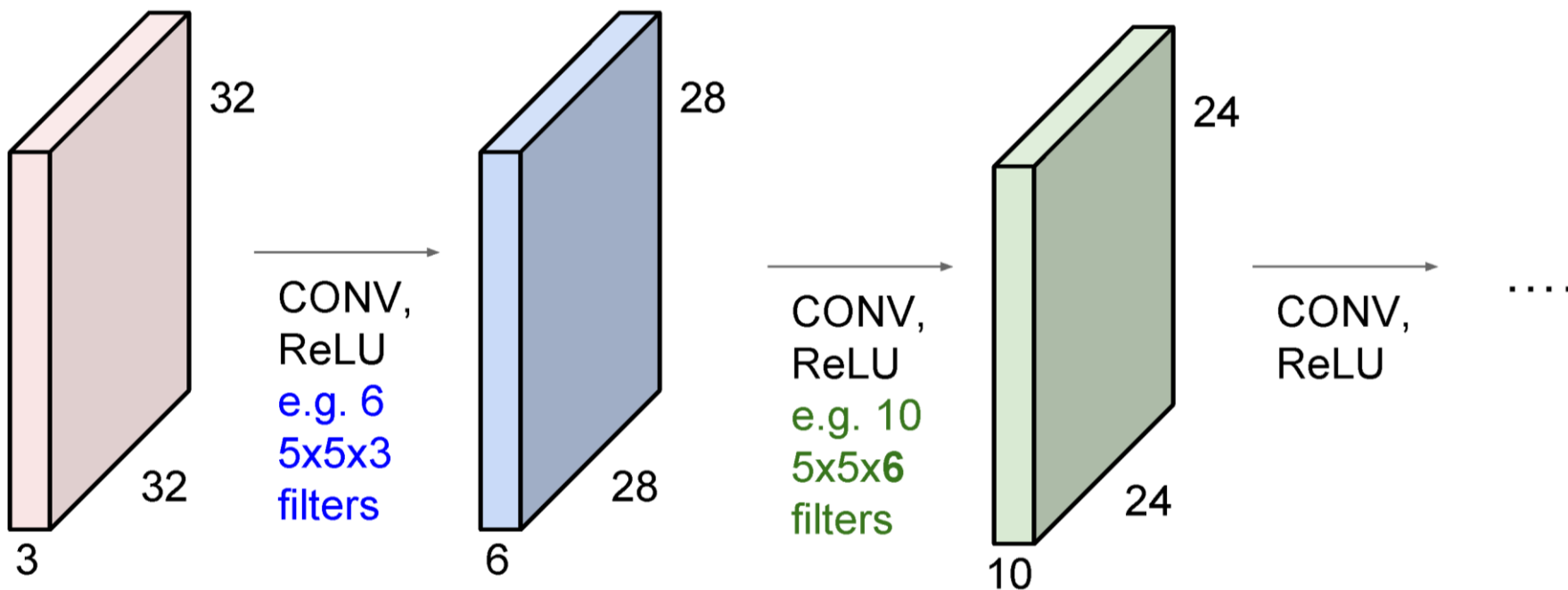
...

...

Convolutional layer input/output



Convolutional layer input/output



Fully-connected vs. Convolutional

Fully-connected layer

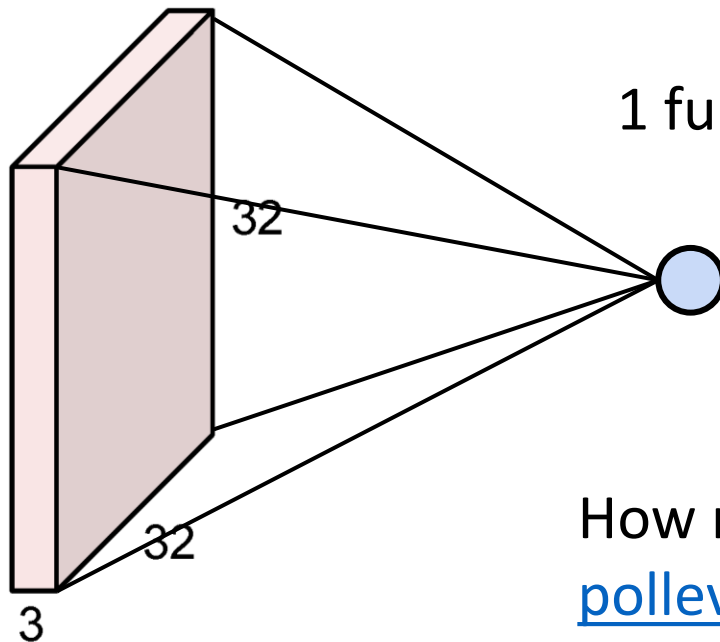
- Each neuron is connected to **every neuron in the input**
- The neuron learns some combination of the input
- The output to the next layer is the neuron's response

Convolutional layer

- Each neuron is connected to a **small patch of the input**
- The neuron learns a convolutional kernel on the input
- The output to the next layer is the input convolved with the neuron's kernel

Fully-connected vs. Convolutional

32x32x3 image



1 fully-connected neuron

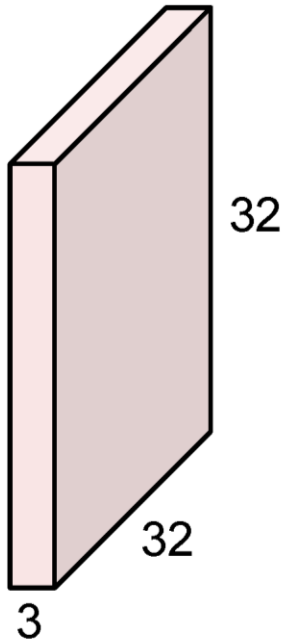
$$32 \times 32 \times 3 + 1(\text{bias}) = 3073$$

How many learned parameters?

pollev.com/krisehinger432

Fully-connected vs. Convolutional

32x32x3 image



1 5x5x3 filter



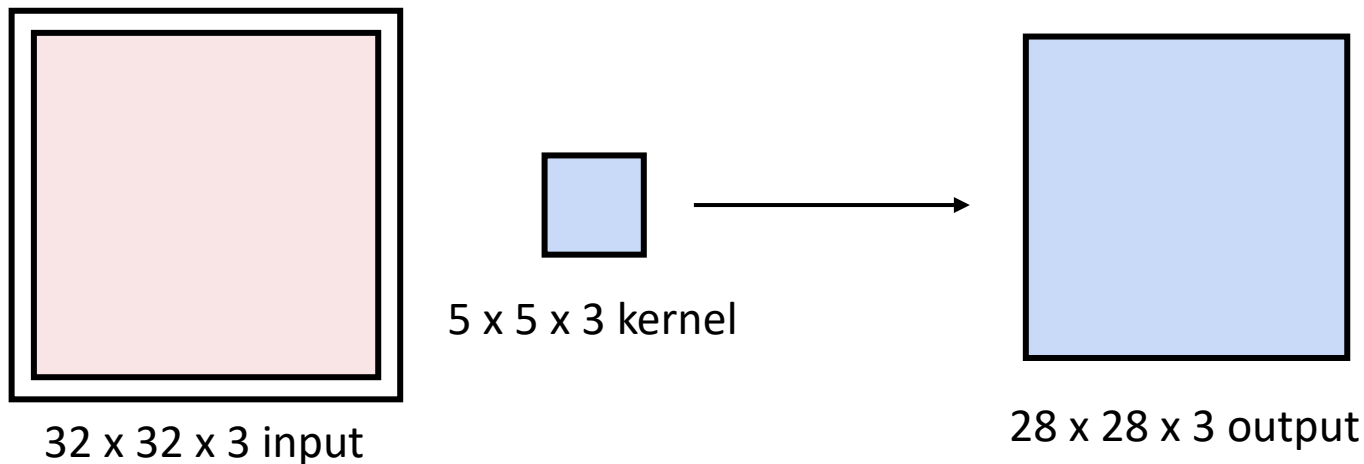
$$5 \times 5 \times 3 + 1 = 76$$

How many learned parameters?

pollev.com/krisehinger432

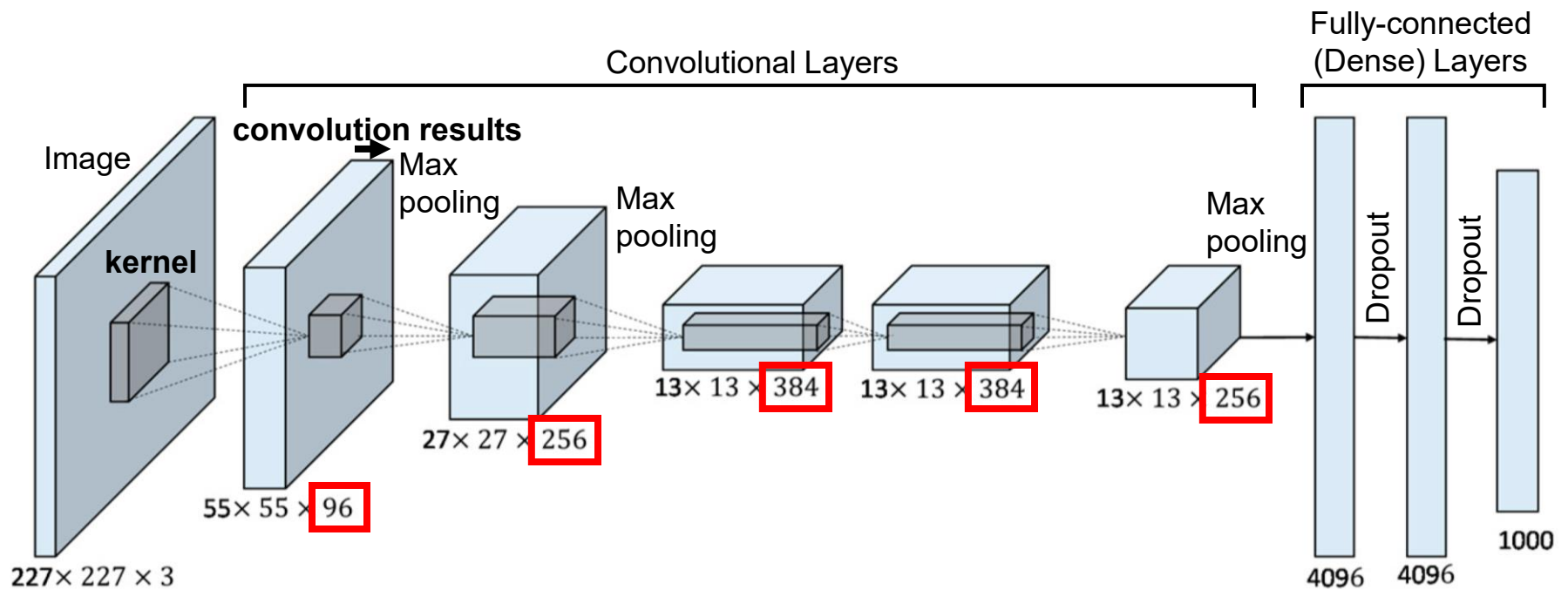
Convolution output size

- **Valid** convolution (with kernel larger than 1x1) results in output smaller than input
- If same-size output is needed, **pad the input** (zero padding is most common)



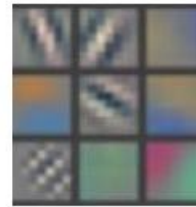
Convolutional neural network

“AlexNet”: Krizhevsky, Sutskever, & Hinton (2012)

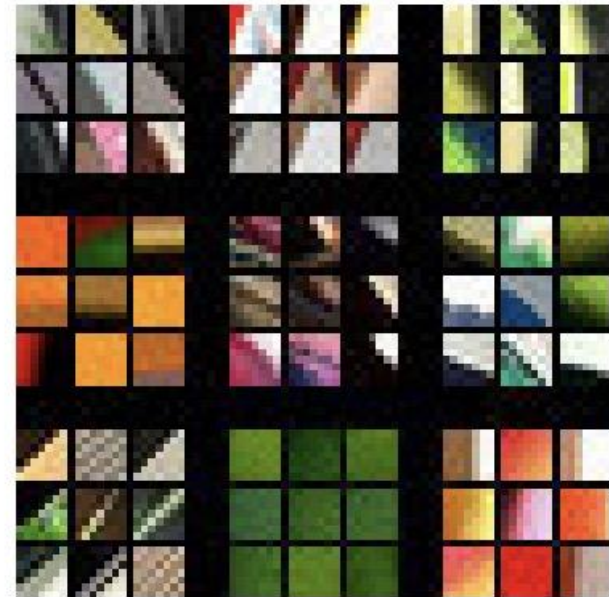


Convolutional layer kernels

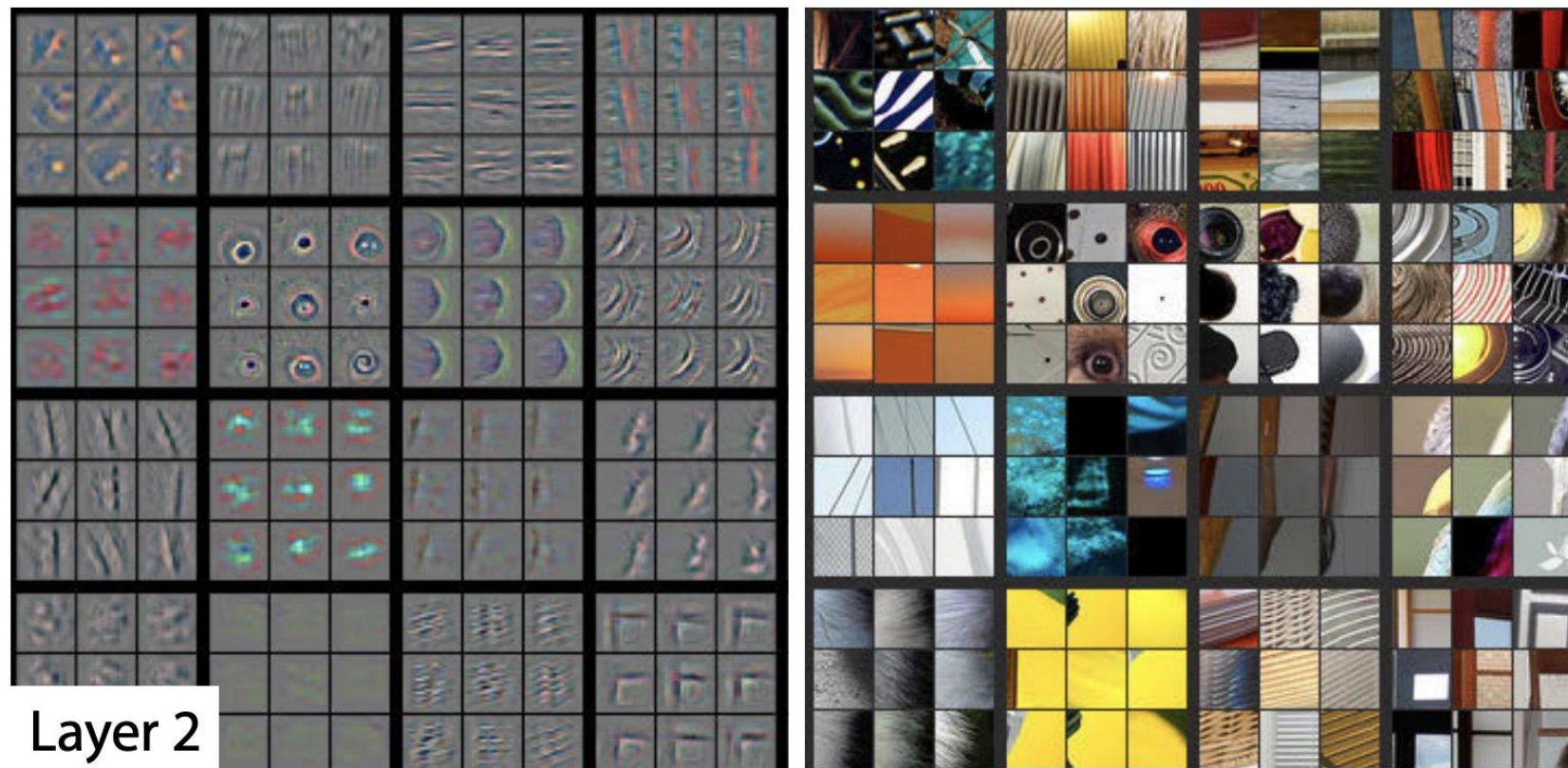
- What kernels do the convolutional layers learn?
- In layer 1, mostly edges
- In higher layers?



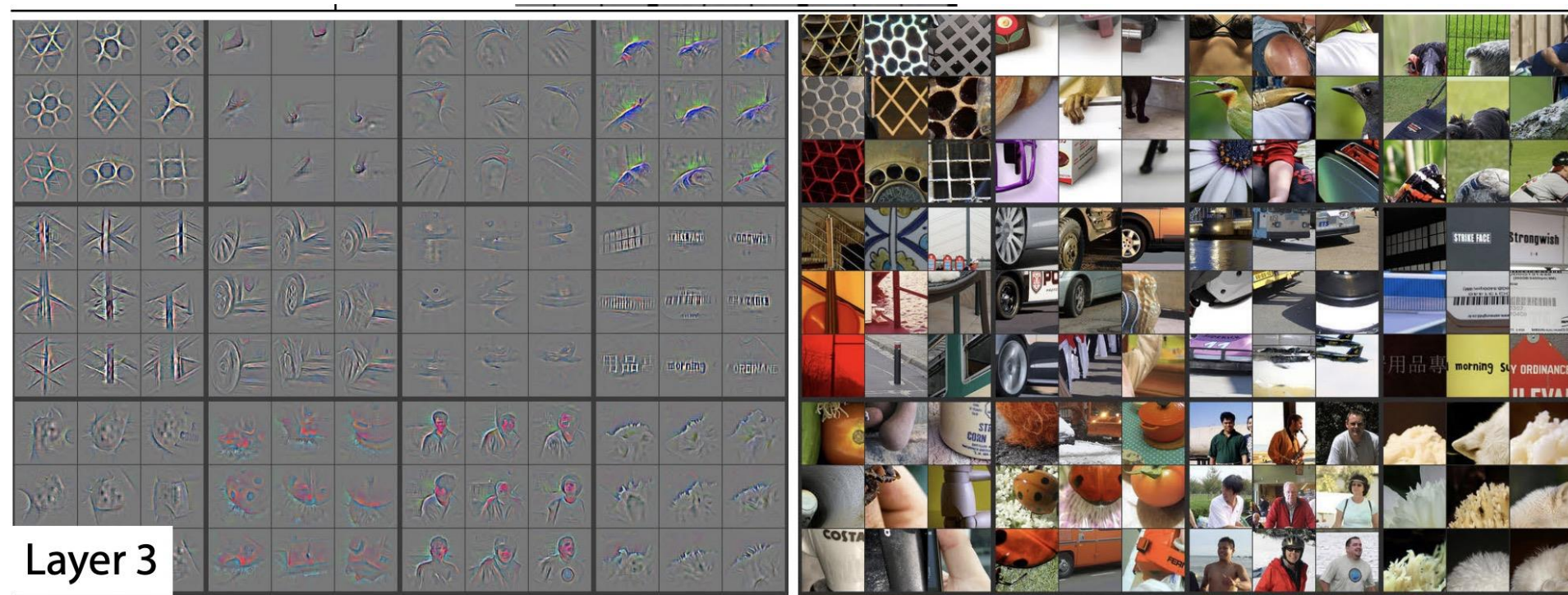
Layer 1



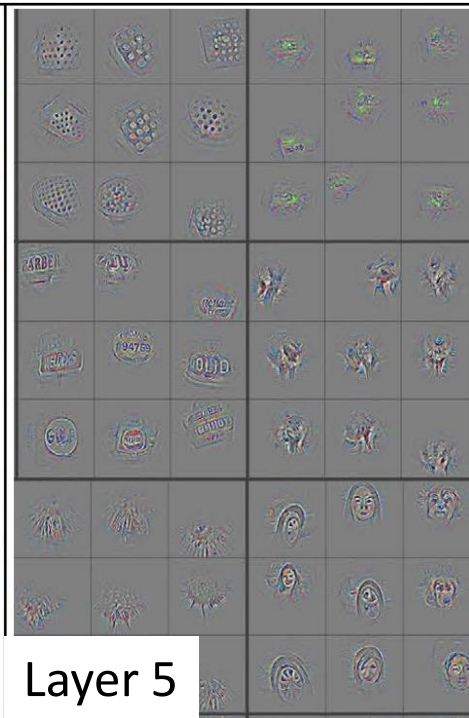
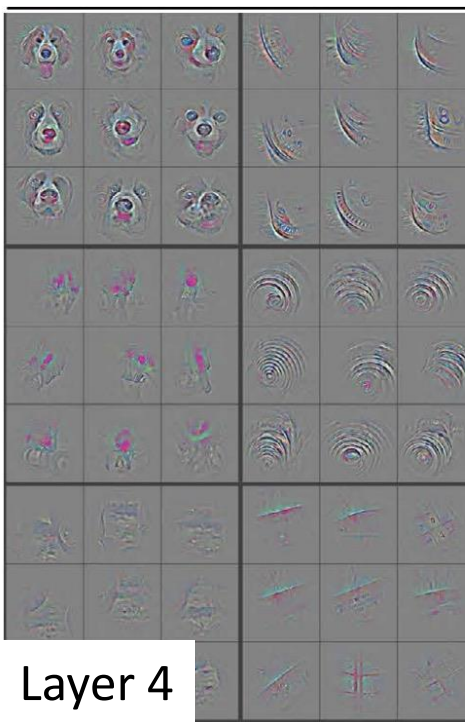
Convolutional layer kernels



Convolutional layer kernels



Convolutional layer kernels



Convolutional layers

- Advantages of convolutional layers
 - Efficient – learns to recognize the same features anywhere in the image, with fewer parameters compared to fully-connected layer
 - Preserves spatial relations – output is an image with values indicating where features are present
- Disadvantages of convolutional layers
 - Limited kernel size means model is limited to learning local features

Summary

- Convolutional neural networks – variation on standard (fully-connected) neural networks
- Each convolutional layer learns a set of kernels and outputs activation maps (= input convolved with learned kernel)